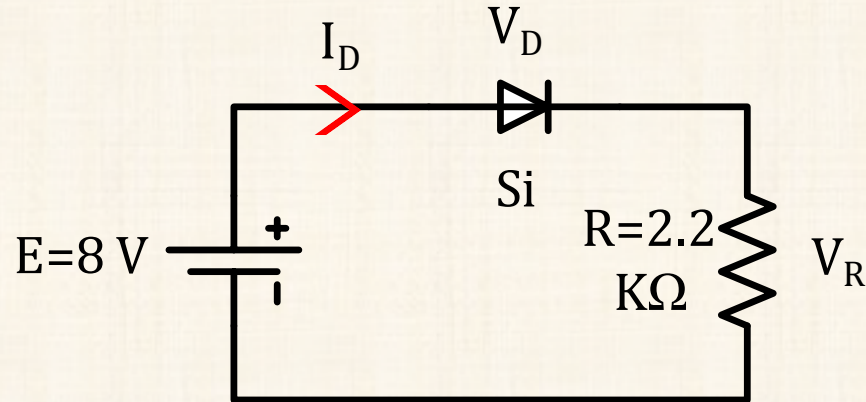


Outline

1. Mathematical Examples

Mathematical Example: 1

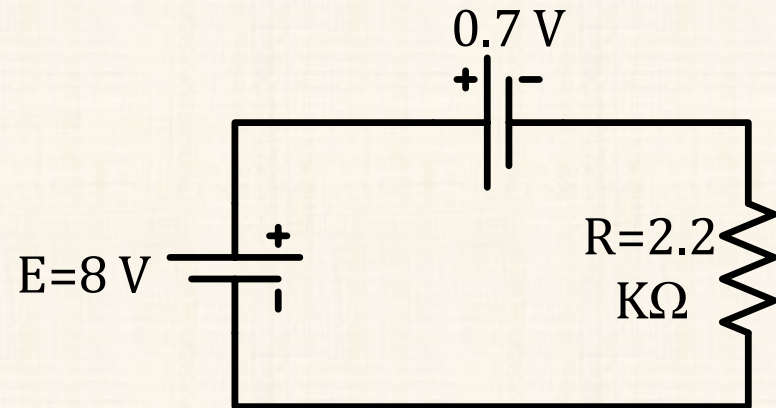


□ Find I_D , V_D and V_R .

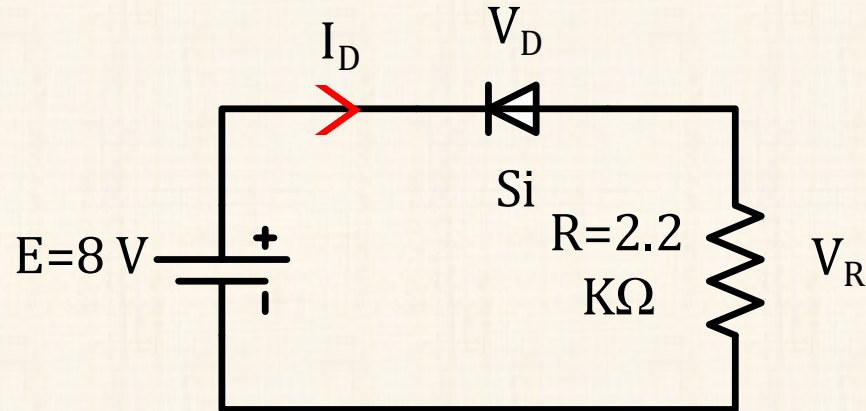
$$V_D = 0.7\text{ V}$$

$$V_R = 8 - 0.7 = 7.3\text{ V}$$

$$I_D = \frac{8 - 0.7}{2.2} = 3.32\text{ mA}$$



Mathematical Example: 2



□ Find I_D , V_D and V_R .

$$-E + V_D + V_R = 0$$

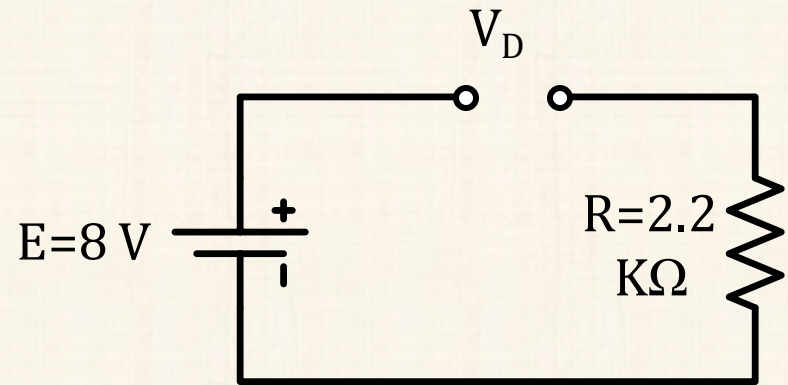
$$\rightarrow V_D = E - V_R$$

$$\rightarrow V_D = E - 0$$

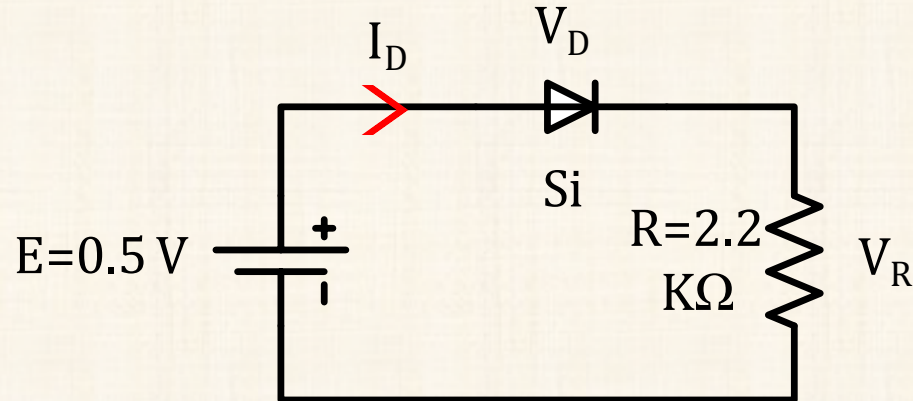
$$\rightarrow V_D = E = 8\text{ V}$$

$$V_R = 0\text{ V}$$

$$I_D = 0\text{ A}$$



Mathematical Example: 3



□ Find I_D , V_D and V_R .

$$-E + V_D + V_R = 0$$

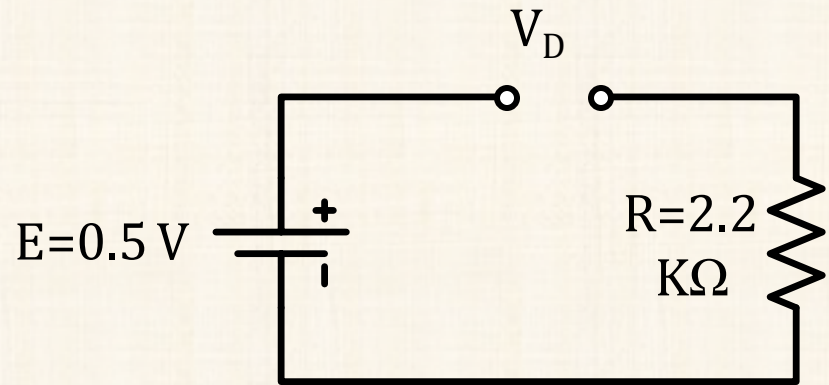
$$\rightarrow V_D = E - V_R$$

$$\rightarrow V_D = E - 0$$

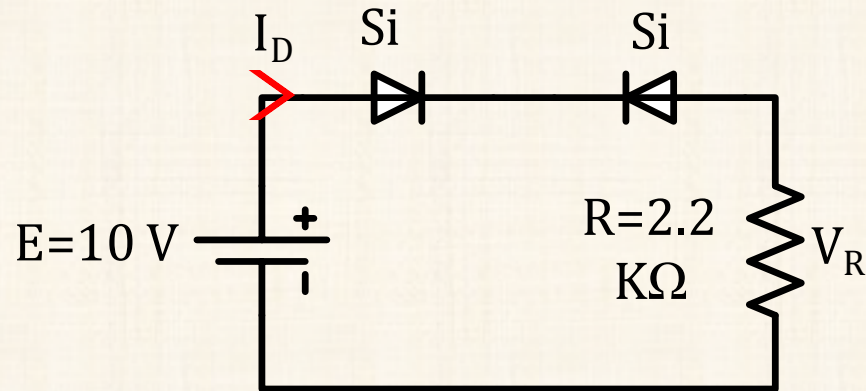
$$\rightarrow V_D = E = 0.5\text{ V}$$

$$V_R = 0\text{ V}$$

$$I_D = 0\text{ A}$$



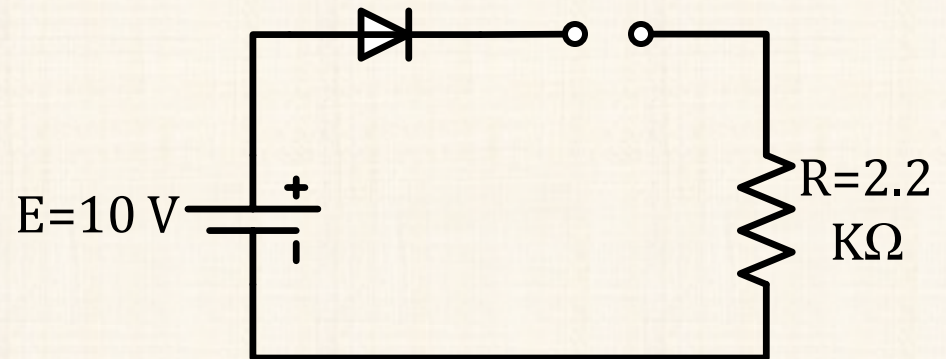
Mathematical Example: 4



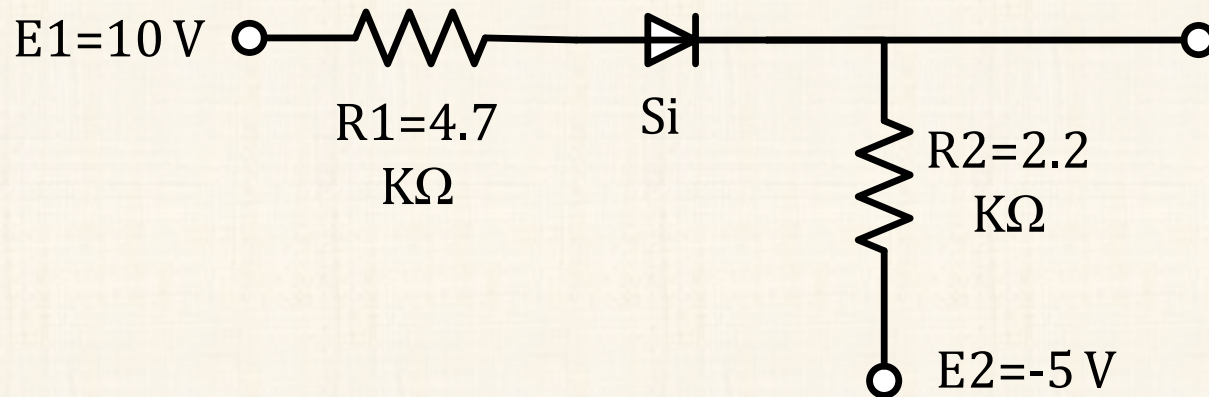
□ Find V_R .

$$I_D = 0\text{ A}$$

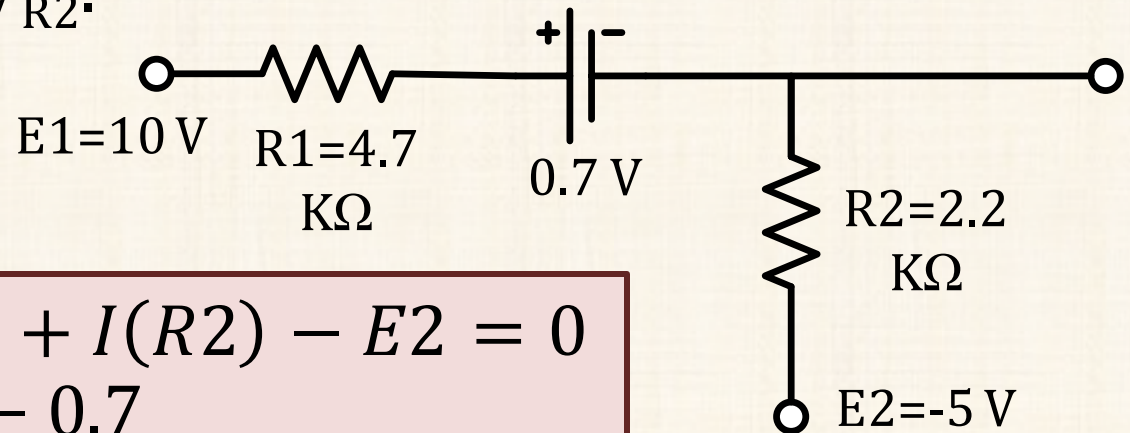
$$V_R = 0\text{ V}$$



Mathematical Example: 5



□ Find V_{R1} and V_{R2} .

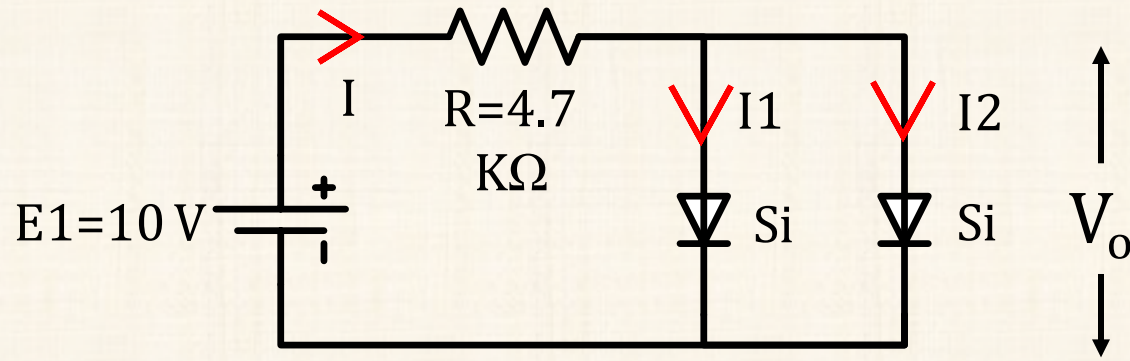


$$-E1 + I(R1) + 0.7 + I(R2) - E2 = 0$$
$$\rightarrow I = \frac{E1 + E2 - 0.7}{R1 + R2} = 2.07 \text{ mA}$$

$$V_{R1} = I(4.7 \text{ k}\Omega) = 9.74 \text{ V}$$

$$V_{R2} = I(2.2 \text{ k}\Omega) = 4.56 \text{ V}$$

Mathematical Example: 6

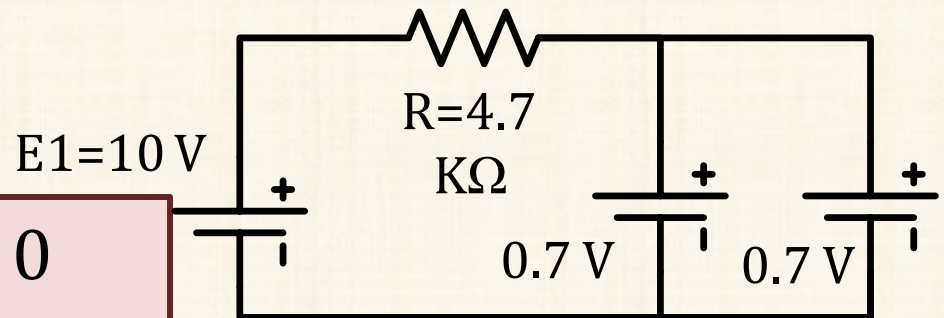


□ Find V_o , I , $I1$ and $I2$.

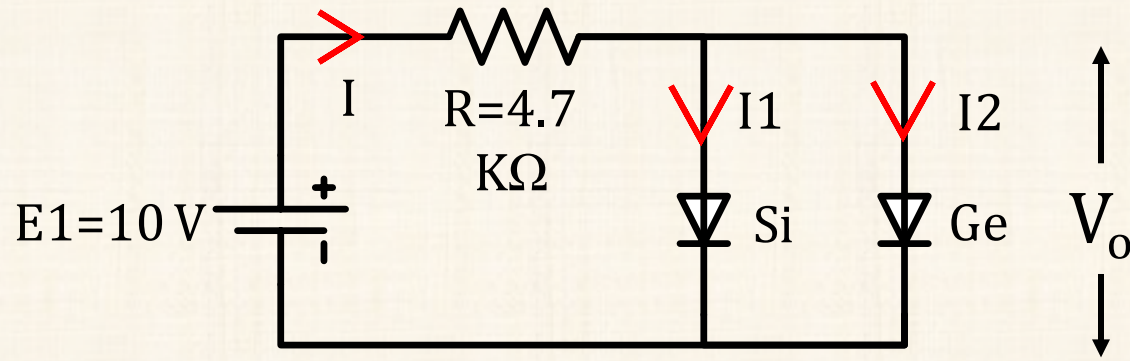
$$V_o = 0.7 \text{ V}$$

$$\begin{aligned} -E1 + I(R) + 0.7 &= 0 \\ \rightarrow I &= \frac{E1 - 0.7}{R} = 1.98 \text{ mA} \end{aligned}$$

$$I1 = I2 = \frac{I}{2} = 0.99 \text{ mA}$$



Mathematical Example: 7

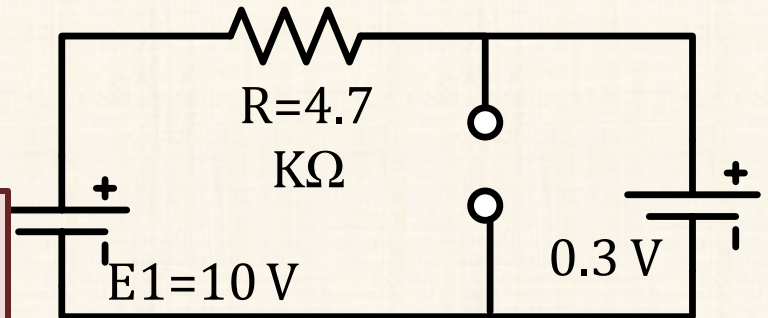


□ Find V_0 , I , I_1 and I_2 .

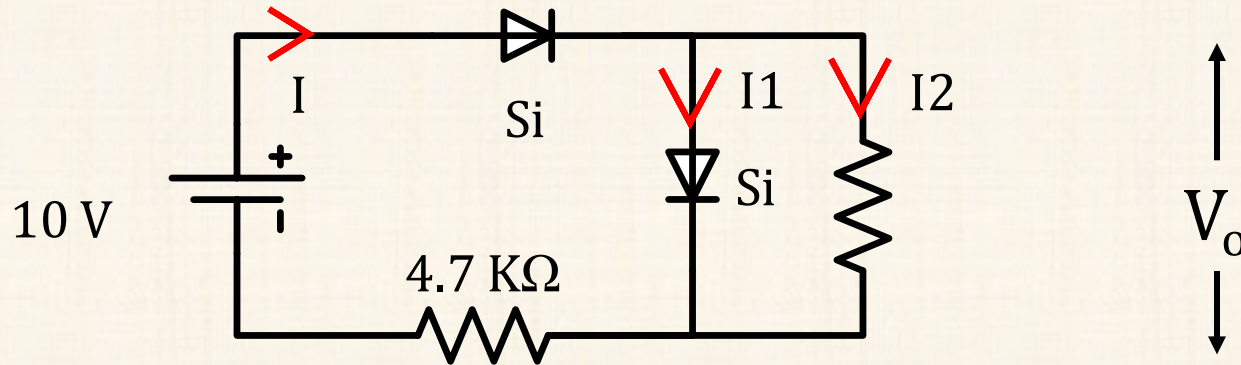
$$V_0 = 0.3 V$$

$$\begin{aligned} -E_1 + I(R) + 0.3 &= 0 \\ \rightarrow I &= \frac{E_1 - 0.3}{R} = 2.06 \text{ mA} \end{aligned}$$

$$I_1 = 0, I_2 = I = 2.06 \text{ mA}$$



Mathematical Example: 8



□ Find V_0 , I , I_1 and I_2 .

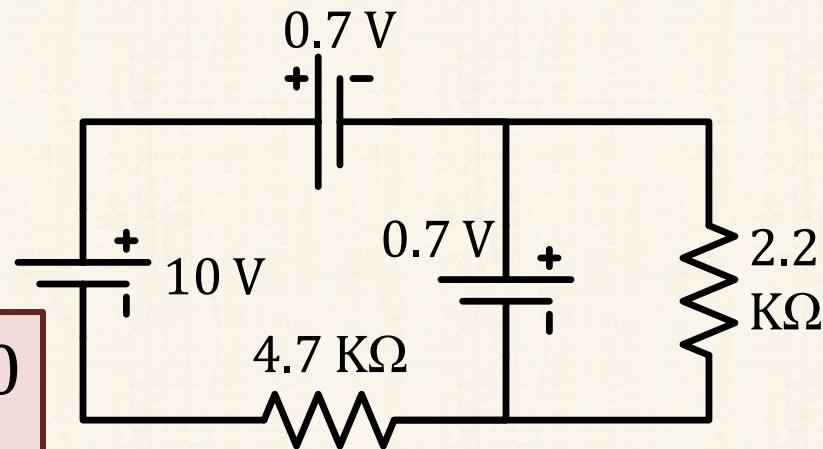
$$V_0 = 0.7 V$$

$$-10 + 0.7 + 0.7 + I(4.7) = 0$$

$$\rightarrow I = \frac{10 - 1.4}{4.7} = 1.83 \text{ mA}$$

$$I_2 = \frac{0.7}{2.2} = 0.32 \text{ mA}$$

$$I_1 = (1.83 - 0.32) = 1.51 \text{ mA}$$



Mathematical Example: 9

- Consider, D_1 and D_2 are ideal diodes. Now Find V and I .

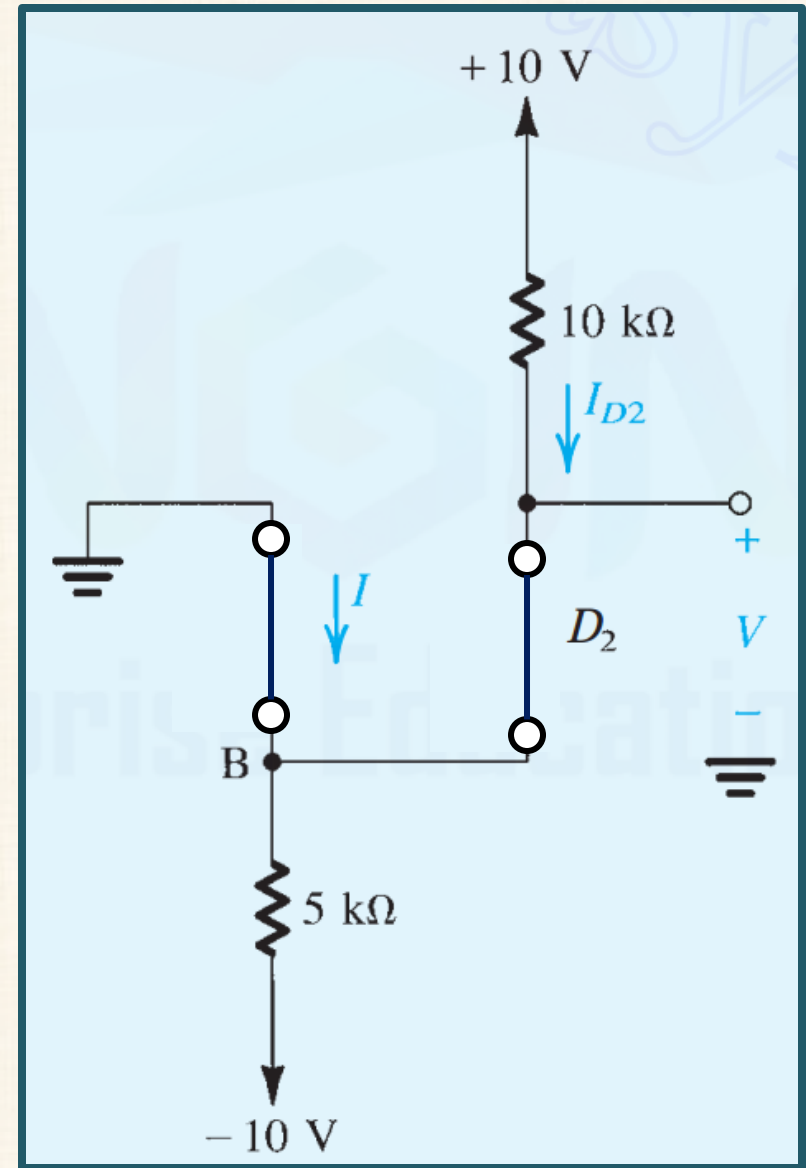
Sol: Let, both diodes are conducting.

So, $V_B = 0\text{ V}$

$$I_{D2} = \frac{10 - 0}{10} = 1\text{ mA}$$

$$I + 1 = \frac{0 - (-10)}{5}$$
$$\rightarrow I = 1\text{ mA}$$

$$V = 0\text{ V}$$



Mathematical Example: 10

➤ Consider, D_1 and D_2 are ideal diodes. Now Find V and I .

Sol: Let, both diodes are conducting.

So, $V_B = 0\text{ V}$

$$I_{D2} = \frac{10 - (-10)}{5 + 10} = 1.33\text{ mA}$$

$$V = (1.33 \times 10) - 10 = 3.33\text{ V}$$

